- November options
- December options
- January options
- February options
- March options
- April options
- May options

but only 3 futures months:

- January future
- March future
- June future

13.3 Underlying Month Selection Rule

Definition 13.2 (Option-to-Future Mapping Rule)**.** An option month is based on the first futures contract that expires after the option itself.

Remark 13.4. This is why nearby option months often share the same underlying futures contract. The option expires first, and the future that comes next in time becomes the underlying contract for that option.

13.4 Example: November Options

Example 13.1 (Which Future Underlies November Options?)**.** Suppose the listed futures months are January, March, and June, while the option months begin with November and December.

November options are not based on a November future, because that future is not listed here. They are instead based on the next futures month in the sequence, which is the **January future**.

13.5 Why This Matters

Remark 13.5. This mapping matters because the option's fair value depends on the price of its underlying future. If you use the wrong futures contract, then your pricing, hedging, and risk calculations will all be off.

Remark 13.6. For market makers, this is one of the first structural details to get right. Before thinking about volatility, edge, or spreads, one must identify the correct underlying contract.

13.6 Connection to Earlier Sections

Remark 13.7. Once the correct underlying future is identified, the trader can use the futures ladder to observe the live bid, ask, and size in that contract. Those prices then feed directly into option valuation and quoting.

14 Payoff Diagrams

Definition 14.1 (Payoff Diagram)**.** A **payoff diagram** is a graph of profit and loss at expiration as a function of the underlying price.

Remark 14.1. In a payoff diagram, the horizontal axis is the underlying price at expiration and the vertical axis is the resulting profit or loss. These diagrams are especially useful for understanding the mechanics of single options and simple option combinations.

Remark 14.2. Although market makers usually do not think in terms of holding one option all the way to expiration, payoff diagrams are still a useful stepping stone because they make the expiration mechanics visually clear.

14.1 Long Underlying at Price 50

Definition 14.2 (Long Underlying)**.** To be **long** the underlying means to have bought it.

Example 14.1. If we buy the underlying at a price of 50, then the payoff at expiration is

$$S_T - 50,$$

where S_T is the underlying price at expiration.

Remark 14.3. This produces a straight 45-degree line. For every \$1 increase in the underlying, profit rises by \$1, and for every \$1 decrease, profit falls by \$1.

14.2 Long Call for Price 1.00

Definition 14.3 (Long Call)**.** A **long call** is the purchase of a call option, giving the right but not the obligation to buy the underlying at the strike price.

Example 14.2. Suppose we buy a call with strike K for a premium of 1.00. Then the payoff at expiration is

$$\max(S_T - K, 0) - 1.00.$$

Remark 14.4. If the underlying finishes below the strike, the call expires worthless and the buyer loses the premium paid. Once the underlying moves above the strike, the call gains value one-for-one with the underlying.

Remark 14.5. For a long call:

- maximum loss $= 1.00$,
- maximum gain $= \infty$,
- breakeven $= K + 1.00$.

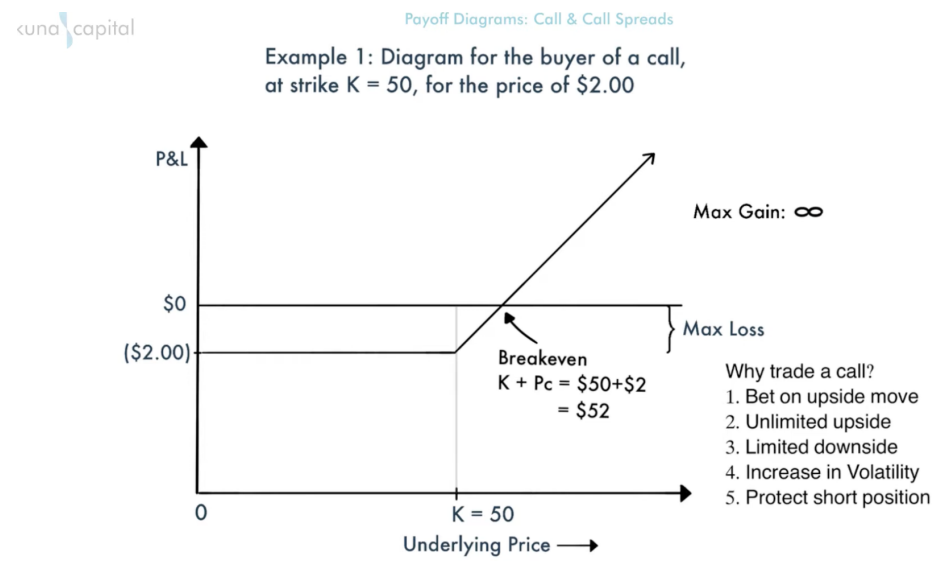

Figure 14: Example payoff diagram for the buyer of a call. Below the strike the option loses only its premium, while above the strike it gains value one-for-one with the underlying.

14.3 Long Put for Price 2.00

Definition 14.4 (Long Put)**.** A **long put** is the purchase of a put option, giving the right but not the obligation to sell the underlying at the strike price.

Example 14.3. Suppose we buy a put with strike K for a premium of 2.00. Then the payoff at expiration is

$$\max(K - S_T, 0) - 2.00.$$

Remark 14.6. If the underlying finishes above the strike, the put expires worthless and the buyer loses the premium paid. If the underlying finishes below the strike, the put gains value one-for-one as the underlying falls.

Remark 14.7. For a long put:

- maximum loss $= 2.00$,
- maximum gain $= K - 2.00$,
- breakeven $= K - 2.00$.

14.4 Long Call Spread for 4.00, Strikes 10 Apart

Definition 14.5 (Long Call Spread)**.** A **long call spread** is created by buying a lower-strike call and selling a higher-strike call.

Example 14.4. Suppose we buy the K_1 call and sell the K_2 call, where

$$K_2 - K_1 = 10,$$

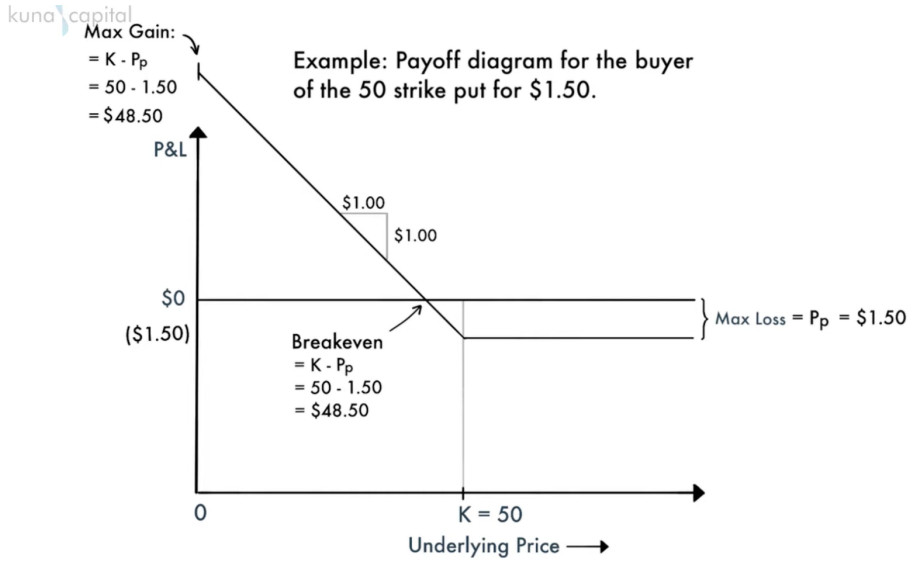

Figure 15: Example payoff diagram for the buyer of a put. Above the strike the option loses only its premium, while below the strike it gains value as the underlying falls.

and we pay 4.00 for the spread. Then the payoff at expiration is

$$\max(S_T - K_1, 0) - \max(S_T - K_2, 0) - 4.00.$$

Remark 14.8. Below K_1, both options expire worthless, so the spread loses its cost of 4.00. Between K_1 and K_2, the lower-strike call gains value. Above K_2, the short higher-strike call offsets any further gain, so the payoff flattens out. This creates the familiar capped-upside spread shape.

Remark 14.9. For a long call spread with width 10 and premium 4.00:

- maximum loss $= 4.00$,
- maximum gain $= 10 - 4.00 = 6.00$,
- breakeven $= K_1 + 4.00$.

14.5 Long Put Spread for 3.00, Strikes 20 Apart

Definition 14.6 (Long Put Spread)**.** A **long put spread** is created by buying a higher-strike put and selling a lower-strike put.

Example 14.5. Suppose we buy the higher-strike put K_1, sell the lower-strike put K_2, where

$$K_1 - K_2 = 20,$$

and pay 3.00 for the spread. Then the payoff at expiration is

$$\max(K_1 - S_T, 0) - \max(K_2 - S_T, 0) - 3.00.$$

Remark 14.10. Above the higher strike, both puts expire worthless and the spread loses its premium. As the underlying falls below the higher strike, the long put begins gaining value. Once

the underlying falls below the lower strike, the short put offsets additional gains, so the diagram flattens on the left.

Remark 14.11. For a long put spread with width 20 and premium 3.00:

- maximum loss $= 3.00$,
- maximum gain $= 20 - 3.00 = 17.00$,
- breakeven $= K_1 - 3.00$.

14.6 Long Call Butterfly for Price 2.00, Strikes 10 Apart

Definition 14.7 (Long Call Butterfly)**.** A **long call butterfly** is formed by buying one lower-strike call, selling two middle-strike calls, and buying one higher-strike call.

Example 14.6. Suppose the strikes are equally spaced:

$$K_1, \quad K_2, \quad K_3$$

with

$$K_2 - K_1 = K_3 - K_2 = 10,$$

and the butterfly costs 2.00. Then the payoff at expiration is

$$\max(S_T - K_1, 0) - 2\max(S_T - K_2, 0) + \max(S_T - K_3, 0) - 2.00.$$

Remark 14.12. A butterfly has a tent-shaped payoff. It loses the premium paid when the underlying finishes too far left or too far right, and makes the most money when the underlying finishes near the middle strike.

Remark 14.13. For a long call butterfly with wing width 10 and premium 2.00:

- maximum loss $= 2.00$,
- maximum gain $= 10 - 2.00 = 8.00$,
- lower breakeven $= K_1 + 2.00$,
- upper breakeven $= K_3 - 2.00$.

Remark 14.14. The seller of the butterfly has the mirror-image payoff about the zero-profit line. The buyer prefers a relatively tight final distribution around the middle strike, while the seller benefits when the underlying moves away from that region.

14.7 Long Straddle for Price 1.00

Definition 14.8 (Long Straddle)**.** A **long straddle** consists of buying a call and a put at the same strike and expiration.

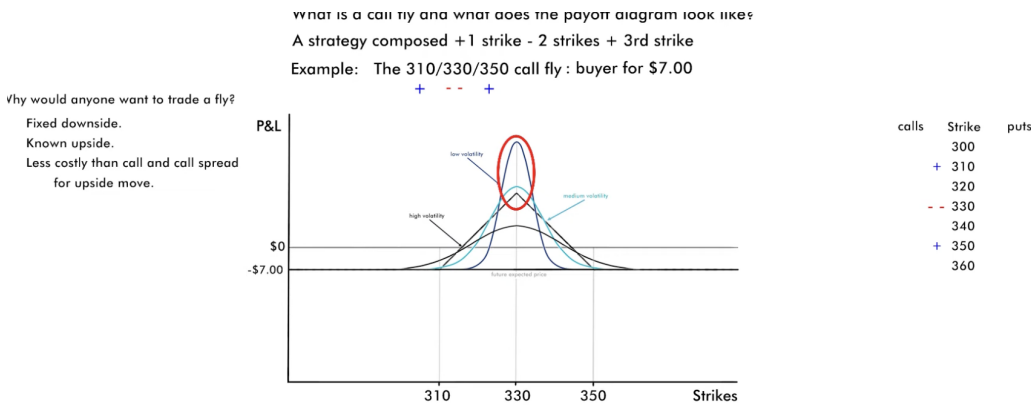

Figure 16: Example payoff diagram for a long call butterfly. The buyer wants the underlying to finish near the middle strike, where the payoff is highest.

Example 14.7. Suppose we buy both the call and the put at strike K, and the total premium paid is 1.00. Then the payoff at expiration is

$$\max(S_T - K, 0) + \max(K - S_T, 0) - 1.00.$$

Remark 14.15. A long straddle profits when the underlying makes a large move in either direction. It loses if the underlying finishes close to the strike, because in that case neither option finishes with enough value to recover the premium paid.

Remark 14.16. For a long straddle purchased for 1.00:

- maximum loss $= 1.00$,
- maximum gain $= \infty$ on the upside and large on the downside down to zero,
- lower breakeven $= K - 1.00$,
- upper breakeven $= K + 1.00$.

14.8 Why Market Makers Use These Only as a Learning Tool

Remark 14.17. On a real options desk, traders usually do not buy one option or one spread and simply hold it to expiration. They trade many options, hedge continuously with the underlying, and manage a large book rather than one isolated payoff diagram. That is why payoff diagrams are best viewed as a mechanical teaching tool rather than a literal picture of how a market-making book is run.

14.9 Basic Options Strategies Explained

Many beginner option strategies can be understood as simple modifications of a basic underlying position. From the simplified novice perspective, these strategies are usually framed around four practical goals:

- generating extra income from a long position,

- protecting an existing long position against downside,
- making a directional bet without owning the underlying outright,
- betting that the underlying will either move a lot or stay near a strike.

Remark 14.18. This section presents these strategies in the simplified way that novice traders often think about them. That framing is useful for intuition, but it ignores many of the risk, pricing, and Greek-related issues that become essential in more serious options trading.

14.9.1 Most People Are Long the Market

A useful starting point is the observation that many investors are already long the market in some form. Individuals buy stocks directly, retirement accounts such as 401(k)s and IRAs often hold stock exposure, and institutions such as pensions and university endowments also tend to own assets that benefit when markets rise.

Remark 14.19. So when we discuss beginner options strategies, many of them are best understood as overlays on top of an already long position in the underlying.

14.9.2 Covered Calls

Definition 14.9 (Covered Call)**.** A **covered call** consists of a long position in the underlying together with a short call option written against that position.

Remark 14.20. The short call brings in option premium immediately. That premium shifts the payoff of the long underlying upward by the amount collected, but in exchange the trader gives up some upside beyond the call strike.

Example 14.8. Suppose an investor owns the underlying at price P and sells a call for 1.00 with strike K. Then:

- the investor receives 1.00 in premium,
- the break-even on the long underlying effectively shifts from P to $P - 1.00$,
- upside past K is capped.

Remark 14.21. This is called a *covered* call because the trader already owns the underlying and can therefore deliver it if the short call is exercised.

Remark 14.22. From the beginner's perspective, a covered call is often viewed as a way to generate "extra cash" if the underlying stays flat, rises only modestly, or even declines slightly.

14.9.3 Puts as Protection

Definition 14.10 (Protective Put)**.** A **protective put** consists of a long position in the underlying together with a long put option.

Remark 14.23. Buying the put costs premium, so it shifts the break-even point against the investor by the amount paid. However, it also limits downside once the underlying falls below the put strike.

Example 14.9. Suppose an investor owns the underlying at price P and buys a put for 1.00 with strike K. Then:

- the investor pays 1.00 out of pocket,
- the break-even shifts from P to $P + 1.00$,
- losses below K are capped.

Remark 14.24. This strategy is commonly described as buying insurance. The investor gives up some immediate cost in order to protect against a severe decline in the underlying.

14.9.4 Directional Trades Without Owning the Underlying

A trader does not need to own the underlying in order to speculate on direction. Options allow directional exposure with a smaller upfront cash outlay than buying the full stock or futures contract.

Bullish view. If a trader thinks the underlying will go up, several simple strategies are common.

Definition 14.11 (Long Call)**.** A **long call** is a bullish trade with limited downside and potentially unlimited upside.

Remark 14.25. The buyer of the call pays the call premium up front. That premium is the known maximum loss.

Definition 14.12 (Long Call Spread)**.** A **long call spread** is a bullish vertical spread formed by buying a lower-strike call and selling a higher-strike call.

Remark 14.26. Compared with a long call, a call spread costs less, but it also caps upside. This makes it a cheaper but more limited bullish trade.

Definition 14.13 (Short Put)**.** A **short put** is another bullish trade because it benefits if the underlying stays above the strike or rises.

Remark 14.27. Although a short put can express a bullish view, it exposes the trader to substantially larger downside risk than simply buying a call.

Bearish view. If a trader thinks the underlying will go down, the natural analogues are:

- buy a put,
- buy a put spread,
- in some cases, sell a call.

Remark 14.28. A long put is the bearish mirror image of a long call. A put spread is the cheaper but capped-upside bearish counterpart. Selling a call also gives downside exposure, though again with more risk than simply buying a put.

14.9.5 Call Spreads and Put Spreads

Definition 14.14 (Call Spread)**.** A **call spread** uses two calls at different strikes to create a capped bullish payoff.

Definition 14.15 (Put Spread)**.** A **put spread** uses two puts at different strikes to create a capped bearish payoff.

Remark 14.29. Spreads are often attractive to novices because they reduce premium cost relative to outright option purchases. The trade-off is that they also reduce the maximum possible gain.

Example 14.10. A trader who buys a 50 strike call for 1.00 and sells a 55 strike call for 0.50 has paid a net premium of

$$1.00 - 0.50 = 0.50.$$

This creates a long call spread with:

- max loss $= 0.50$,
- max gain $= 5.00 - 0.50 = 4.50$,
- breakeven $= 50.50$.

14.9.6 Straddles

Definition 14.16 (Straddle)**.** A **straddle** consists of buying a call and a put at the same strike and expiration.

Remark 14.30. A long straddle is a bet on movement, not direction. The trader wants the underlying to move far enough away from the strike that one side of the position more than pays for the total premium spent.

Remark 14.31. This is useful when the trader believes the underlying will make a large move but does not know whether that move will be upward or downward.

Example 14.11. Suppose a trader buys a call for 1.00 and a put for 0.80 at the same strike. Then the total premium paid is

$$1.00 + 0.80 = 1.80.$$

The trader profits only if the underlying moves far enough away from the strike to overcome that 1.80 cost.

Definition 14.17 (Short Straddle)**.** A **short straddle** consists of selling both the call and the put at the same strike.

Remark 14.32. The short straddle is the opposite bet. Instead of wanting the underlying to move a lot, the trader wants it to stay near the strike so that both options expire with little or no value.

Example 14.12. If the trader sells the call for 1.00 and the put for 0.80, then the total premium collected is

$$1.80.$$

That 1.80 is the maximum profit, and it occurs if the underlying finishes exactly at the strike at expiration.

Remark 14.33. The danger of the short straddle is that large moves in either direction can create very large losses.

14.9.7 A Simplified Summary

From the novice perspective, the most common simple strategies can be summarized as follows:

- **Covered call:** collect premium, but cap upside.
- **Protective put:** pay premium, but limit downside.
- **Long call / long put:** simple directional bets with known premium loss.
- **Call spread / put spread:** cheaper directional bets with capped gains.
- **Straddle:** a bet that the underlying will move a lot.

Remark 14.34. These are the "paper" strategies most commonly discussed in popular options content. Later, with Greeks and more complete pricing tools, the same strategies can be analyzed much more rigorously in terms of volatility, convexity, time decay, and hedging.

14.10 Time Premium and Put–Call Parity

Option prices can be broken into two parts:

- **Intrinsic value**, and
- **Extrinsic value**, also called **time premium**.

This decomposition is one of the cleanest ways to understand how options are priced.

14.10.1 In-the-Money and Out-of-the-Money

Definition 14.18 (In-the-Money)**.** An option is **in-the-money (ITM)** if it already has value based on the current relationship between the underlying price and the strike.

Definition 14.19 (Out-of-the-Money)**.** An option is **out-of-the-money (OTM)** if it has no intrinsic value at the current underlying price.

For calls and puts:

- a call is ITM when $U > K$,
- a call is OTM when $U < K$,
- a put is ITM when $K > U$,
- a put is OTM when $K < U$.

Remark 14.35. So calls with strike below the underlying are ITM, while puts with strike above the underlying are ITM. The other two regions are OTM.

14.10.2 Intrinsic Value

Definition 14.20 (Intrinsic Value)**.** The **intrinsic value** of an option is the amount the option is worth from immediate exercise value alone.

For calls,

$$\text{Intrinsic Value(call)} = \max(0, U - K).$$

For puts,

$$\text{Intrinsic Value(put)} = \max(0, K - U).$$

Example 14.13. If the underlying is trading at 370 and we look at a 330 call, then the call has intrinsic value

$$370 - 330 = 40.$$

If instead we look at a 430 put, then the put has intrinsic value

$$430 - 370 = 60.$$

Remark 14.36. An ITM option must be worth at least its intrinsic value. An OTM option has zero intrinsic value.

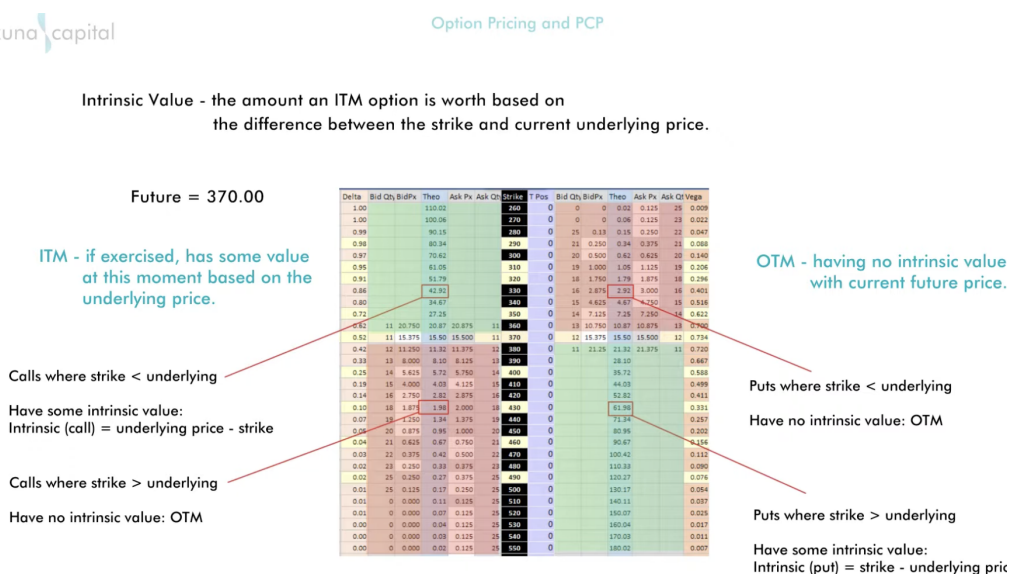

Figure 17: Intrinsic value across strikes. Calls with strike below the underlying have intrinsic value equal to $U - K$, while puts with strike above the underlying have intrinsic value equal to $K - U$.

14.10.3 Extrinsic Value or Time Premium

Definition 14.21 (Extrinsic Value / Time Premium)**.** The **extrinsic value**, also called **time premium**, is the part of the option price that comes from optionality rather than immediate exercise.

So every option price satisfies

$$\text{Option Price} = \text{Intrinsic Value} + \text{Extrinsic Value}.$$

This means:

- ITM options have both intrinsic and extrinsic value,
- OTM options have only extrinsic value.

Example 14.14. Suppose the underlying is 370 and a 330 call trades at 42.92. Its intrinsic value is

$$370 - 330 = 40,$$

so its extrinsic value is

$$42.92 - 40 = 2.92.$$

Remark 14.37. Extrinsic value exists because the contract still has time left. Even an OTM option can become valuable before expiration if the underlying moves enough.

Remark 14.38. The three main drivers of extrinsic value are:

- distance to at-the-money,
- time to expiration,
- implied volatility.

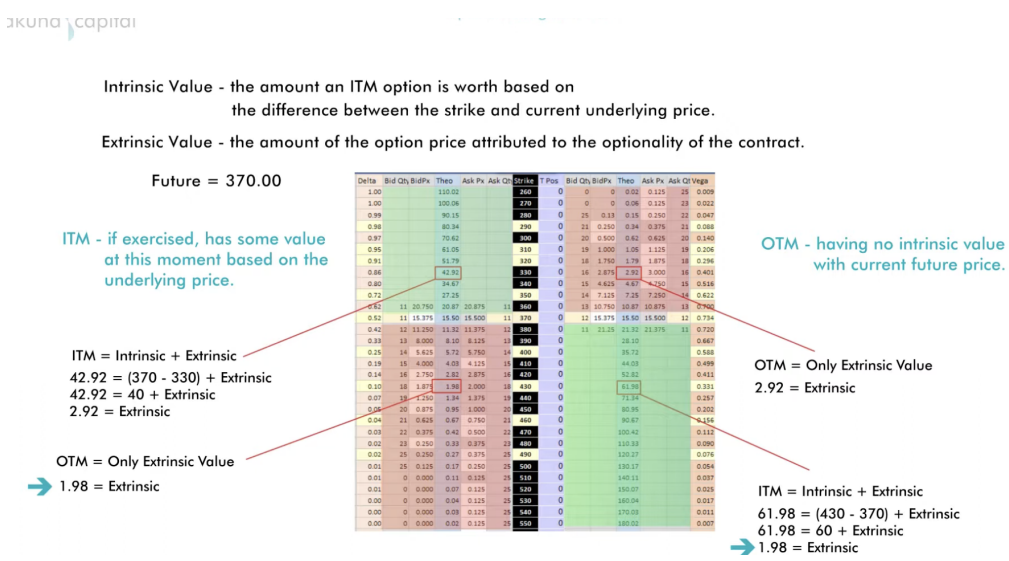

Figure 18: Option price decomposed into intrinsic and extrinsic value. OTM options contain only extrinsic value, while ITM options contain both intrinsic and extrinsic value.

14.10.4 Forward Price Intuition

Before discussing put–call parity, it is useful to think in terms of the **forward price** of the underlying rather than just the current spot price. The forward reflects the current underlying level together with interest rates, dividends, and carrying costs when relevant.

Remark 14.39. Once the forward price is fixed, the strike's distance from that forward is what determines whether the option is ITM or OTM and how much intrinsic value it has.

14.10.5 Put–Call Parity

Definition 14.22 (Put–Call Parity)**.** For a call and put on the same strike and expiration, the simplified put–call parity relation is

$$C - P = U - K.$$

Equivalently,

$$U = C - P + K.$$

This identity is fundamental. If it did not hold, an arbitrage opportunity would exist.

Remark 14.40. Put–call parity says that a call and put on the same line are mathematically linked. Knowing any three of the quantities C, P, U, and K determines the fourth.

14.10.6 Synthetic Underlying Interpretation

A very useful interpretation is:

$$\text{long call} + \text{short put} \equiv \text{long underlying forward}.$$

Why? At expiration:

- if the underlying finishes above the strike, the call is exercised,
- if the underlying finishes below the strike, the short put is exercised against you.

Either way, you end up effectively buying the underlying at the strike price.
Similarly,

$$\text{short call} + \text{long put} \equiv \text{short underlying forward}.$$

Remark 14.41. This synthetic interpretation is the intuition behind PCP. Calls, puts, and the underlying are not independent securities. They can be transformed into one another through replication.

14.10.7 A Pricing Example

Example 14.15. Suppose

$$C = 15.34, \qquad U = 85, \qquad K = 75.$$

Use put–call parity:

$$C - P = U - K.$$

So

$$15.34 - P = 85 - 75 = 10.$$

Hence

$$P = 15.34 - 10 = 5.34.$$

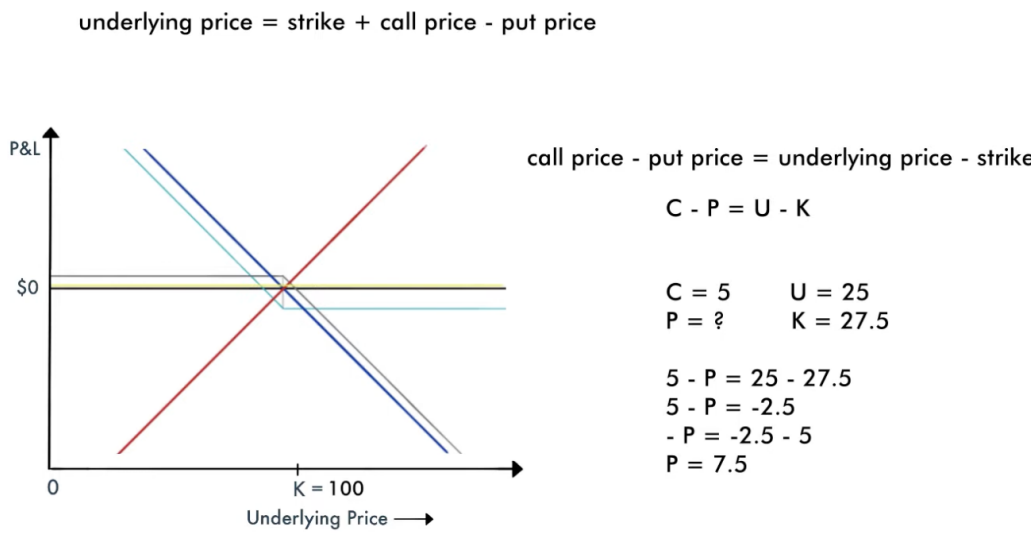

Figure 19: Put–call parity viewed through payoff replication. The relationship $C - P = U - K$ links call price, put price, underlying price, and strike so that no arbitrage exists.

Now break the call into intrinsic and extrinsic parts:

$$\text{Intrinsic(call)} = 85 - 75 = 10,$$

so the call's extrinsic value is

$$15.34 - 10 = 5.34.$$

Remark 14.42. In this example, the call consists of 10 of intrinsic value and 5.34 of extrinsic value. The corresponding put value 5.34 is the matching time premium in the simplified zero-interest interpretation.

14.10.8 Interest Rates and the Forward Version

The simplified formula

$$C - P = U - K$$

is best for intuition. In practice, financing matters, so traders often use a forward-adjusted version:

$$F = (C - P)e^{rt} + K,$$

where F is the forward price, r is the interest rate, and t is time to expiration.

Remark 14.43. Across strikes, raw values of $C - P + K$ may not imply exactly the same underlying unless interest rates are incorporated. Once financing is handled properly, the implied forward values line up much more closely.

14.11 Video: Lecture 2 — Theos and Combination Spreads

This lecture focuses on two ideas that are central to practical options market making:

1. **theoretical values** (or *theos*), and

2. **combination spreads**, also called *combos*.

14.11.1 Theoretical Values and Edge

A market maker constantly keeps track of a theoretical fair value for each option. This theoretical value is called the **theo**. The point of the theo is not just to estimate what an option is worth, but to measure whether a trade was favorable or unfavorable relative to the market maker's own pricing model.

Definition 14.23 (Theoretical Value)**.** The **theoretical value** of an option is the market maker's internal estimate of the fair price of that contract.

Remark 14.44. Unlike a passive investor, a market maker actively updates theos as inputs move. In practice, this often means adjusting volatility assumptions, especially implied volatility.

The lecture emphasizes that market makers care deeply about controlling their theos because that is how they track:

- whether they bought cheap or sold rich,
- how much edge they captured,
- and how their book's valuation changes as the market moves.

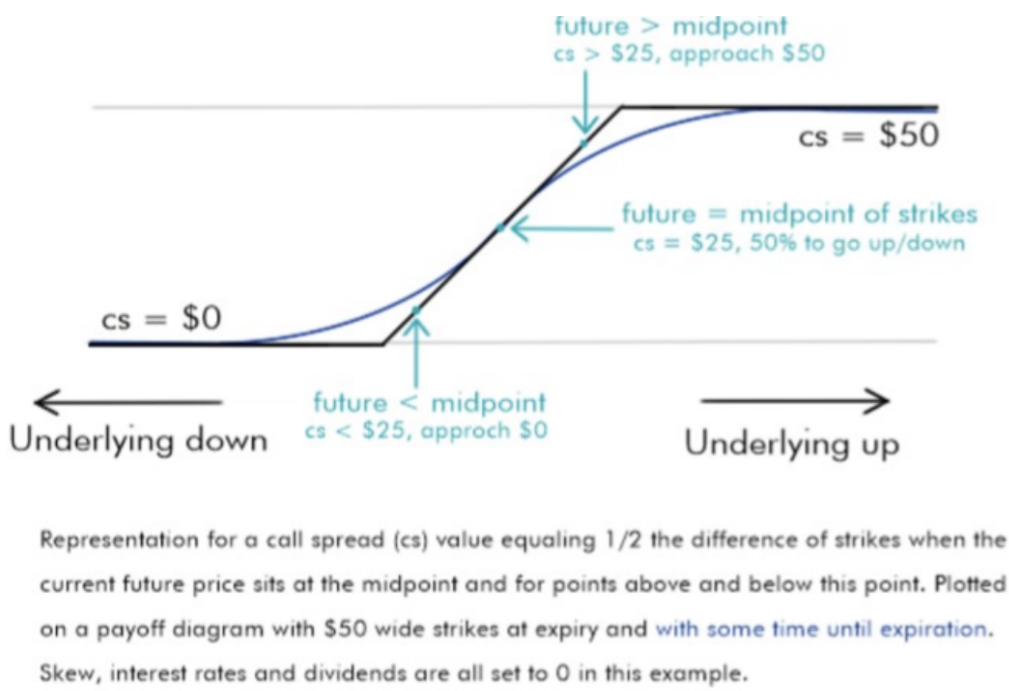

Figure 20: Reviewing the role of theoretical values. Market makers control their theos by adjusting model inputs such as implied volatility.

14.11.2 The Long-Run Anchor of Option Prices

One important intuition from the lecture is that option prices do not float freely forever. As expiration approaches, an option must converge toward either:

- 0, if it expires out of the money, or
- its **intrinsic value**, if it expires in the money.

Remark 14.45. This is a useful anchor for thinking about theos. Whatever the current model price is, at expiration the optionality disappears. What remains is only exercise value.

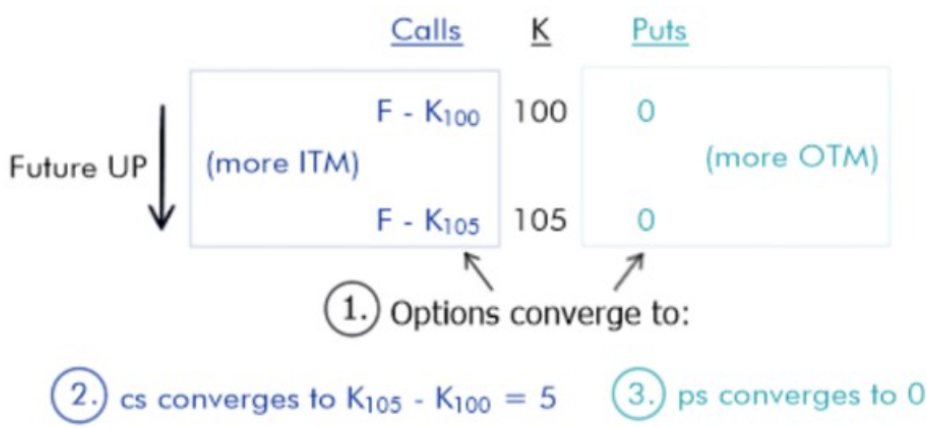

Figure 21: Option value above intrinsic prior to expiration. The gap between the market value curve and intrinsic value represents time value.

14.12 Options Limits and Boundaries

There are several structural relationships in options that should always hold. These bounds are useful both for checking whether a quoted market is reasonable and for identifying possible mispricings. They are not modeling assumptions. They follow from the contract payoffs themselves.

14.12.1 Basic Bounds

Some of the most important option limits are:

- An option can never be worth less than 0.
- An in-the-money call can never be worth less than its intrinsic value:

$$\max(F - K, 0).$$

- An in-the-money put can never be worth less than its intrinsic value:

$$\max(K - F, 0).$$

- A call spread can never be worth less than 0 and can never be worth more than the strike difference.
- A put spread can never be worth less than 0 and can never be worth more than the strike difference.
- A symmetric butterfly can never be worth less than 0.
- A same-strike calendar spread on the same underlying should not be worth less than 0, since the longer-dated option contains at least as much optionality as the shorter-dated one.
- Put–call parity should hold. If it does not, there is an arbitrage inconsistency.

Remark 14.46. These bounds are some of the fastest and most useful sanity checks in options trading. Before doing anything fancy, a trader should first ask whether a quoted value even satisfies the basic payoff constraints.

14.12.2 Intrinsic Value as a Hard Lower Bound

If a call is already in the money, it must be worth at least the amount that could be realized by immediate exercise. Likewise for a put. This gives the lower bounds

$$C \geq \max(F - K, 0), \qquad P \geq \max(K - F, 0).$$

So if the future is above the strike, the call must be worth at least $F - K$. If the future is below the strike, the put must be worth at least $K - F$.

Remark 14.47. An ITM option may be worth more than intrinsic value because it still has time value, but it should never be worth less.

14.12.3 Call Spread Bounds

Consider a call spread with strikes $K_1 < K_2$, defined by

$$\mathrm{CS} = C(K_1) - C(K_2).$$

At expiration, this spread has only three possible regimes:

- If $F \leq K_1$, both calls expire worthless, so

$$\mathrm{CS} = 0.$$

- If $K_1 < F < K_2$, the lower-strike call is in the money and the upper-strike call is not, so

$$\mathrm{CS} = F - K_1.$$

- If $F \geq K_2$, both calls are in the money, and their difference is fixed:

$$\mathrm{CS} = (F - K_1) - (F - K_2) = K_2 - K_1.$$

Therefore,

$$0 \leq \text{CS} \leq K_2 - K_1.$$

This is why a call spread can never be negative and can never be worth more than the width between strikes.

14.12.4 Midpoint Approximation for a Call Spread

A very useful rule of thumb is that if the future is roughly at the midpoint of the two strikes, then the call spread should be worth roughly half the strike width.

For example, if the strikes are 100 and 150, then the width is 50. If the future is around the midpoint 125, then a reasonable first approximation is

$$\text{CS} \approx 25.$$

The intuition is simple:

- far below the lower strike, the spread should behave like 0,
- far above the upper strike, the spread should behave like the full strike width,
- near the midpoint, it should sit around half-width.

Remark 14.48. With time still remaining, the spread transitions smoothly between these extremes rather than jumping sharply. But the midpoint approximation is still a very useful mental anchor.

14.12.5 How the Individual Legs Converge

Another way to understand spread limits is to reason directly from the individual options as expiration approaches.

Suppose the future is above both strikes $K_{100} = 100$ and $K_{105} = 105$. Then the two calls converge to intrinsic value:

$$C(100) \to F - 100, \qquad C(105) \to F - 105.$$

So the call spread converges to

$$(F - 100) - (F - 105) = 5.$$

Thus the 100/105 call spread converges to its full width, namely 5.

Now look at the puts with those same strikes. If the future is above both strikes, both puts are out of the money, so each converges to 0. Hence the corresponding put spread converges to 0.

This gives a clean rule:

- when the future finishes above both strikes, the call spread converges to full width and the put spread converges to 0,
- when the future finishes below both strikes, the put spread converges to full width and the call spread converges to 0.